XIAMEN UNIVERSITY MALAYSIA
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EEE312 Week 13

Implementation of Lagrange Interpolation in C Programming Language

Apply the Lagrange interpolation method to interpolate the following sets of points:

{(0, 3), (1, 5), (4, -1)}
{(0, 3), (1, 5), (2, -1), (3, 0)}

For the Lagrange polynomials derived from the points above, find the value of $P(1.5)$.

Beginning of C Code

```
1  // Lagrange interpolation
2  #include <stdio.h>
3
4  int main()
5  {
6      float x[3] = {0, 1, 4};
7      float y[3] = {3, 5, -1};
8      float xi = 1.5;
9      float a[3]; // coefficients a0, a1 and a2
10     float P;    // polynomial P(x)
```

Line 6-7: Declare and initialize the points (x, y).
Line 8: Declare and initialize the point xi= 1.5. for finding the interpolate point P(xi).
Line 9: Declare the variables a[0], a[1] and a[2] to store the constants of polynomial P(x).
Line 10: Declare the variable Pto store the value of P(xi).

```
12      a[0] = y[0]/((x[0]-x[1])*(x[0]-x[2]));
13      a[1] = y[1]/((x[1]-x[0])*(x[1]-x[2]));
14      a[2] = y[2]/((x[2]-x[0])*(x[2]-x[1]));
15
16      P = a[0]*(xi-x[1])*(xi-x[2])+a[1]*(xi-x[0])*(xi-x[2])+a[2]*(xi-x[0])*(xi-x[1]);
17
18      for(int i=0; i<3; i++)
19          printf("\na[%d]=%f.", i, a[i]);
20
21      printf("\nP(%.1f) = %f.\n", xi, P);
22
23      return 0;
24  }
```

Line 12: Find a[0]using the equation

$$a_0 = \frac{y_0}{(x_0 - x_1)(x_0 - x_2)}$$

Line 13: Find a[1]using the equation

$$a_1 = \frac{y_1}{(x_1 - x_0)(x_1 - x_2)}$$

Line 14: Find a[2]using the equation

$$a_2 = \frac{y_2}{(x_2 - x_0)(x_2 - x_1)}$$

Line 16: Find P(x) where x= 1.5 using the equation

$$P_2(x) = a_0(x - x_1)(x - x_2) + a_1(x - x_0)(x - x_2) + a_2(x - x_0)(x - x_1),$$

Line 18-19: Print the values of a[0], a[1]and a[2]at the program output.
Line 21: Print the values of the P(1.5)at the program output.

## Output


a[0]=0.750000.
a[1]=-1.666667.
a[2]=-0.083333.
P(1.5) = 5.250000.
Process returned 0 (0x0)   execution time : 3.060 s
Press any key to continue.